Pareto Improving Fiscal and Monetary Policies: Samuelson in the New Keynesian Model

Discussion of Aguiar, Amador & Arellano (2024)

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What the Paper Does

An elegant Samuelson-meets-NK model to study Pareto efficiency with heterogeneity

- Perpetual youth OLG (Blanchard–Yaari) + textbook NK (Rotemberg pricing)
- Workers save in govt bonds; entrepreneurs hold equity (segmentation)
- Non-Ricardian: govt bonds are net wealth for workers

Core Question

Can debt-financed transfers be Pareto improving? Under what monetary rule?

Why is this hard? efficient for whom is non-trivial even in a constant-K world (eg Aguiar et al 2024).

The Model: Two Equations

Equilibrium reduces to two ODEs in worker consumption c and inflation π:

Euler equation:

$$\dot{c} = (r - \rho - g + \alpha)\, c \; - \; \mu\, b$$

$-\mu b$: dying agents hold wealth, newborns hold zero. Demographic turnover: dying old consume more than new-born. Source of non-Ricardian effects.

Phillips curve:

$$\dot{\pi} = \hat{\rho}\, \pi + \kappa\, (c^* - c)$$

When $c > c^*$: wages above flex-price level $\rightarrow$ markups compressed $\rightarrow$ inflation rises.

Output is exogenous: policy changes *distribution*, not output.

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Real debt: sidestep FTPL type fiscal monetary interactions

Alternative tractable micro-foundation

Caramp and Singh (2024) retirement risk with imperfect insurance yields similar Euler:
CS Generalized Euler Equation:

$$\frac{1}{C_t^w} = \beta(1+r_t)\left[\underbrace{\frac{1-\chi}{C_{t+1}^w}}_{\text{stay worker}} + \underbrace{\frac{\chi}{B_{t+1}+w^o}}_{\text{retire}}\right]$$

- Standard Euler: consumption smoothing if remain worker
- Retirement motive: bonds are consumption source in retirement
- Bond premium: $bp = \chi\left(C^w/(B+w^o)-1\right) > 0$ when bonds scarce

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	AAA	**CS**
Wedge	$-\mu B$ (demographic turnover)	$\chi/(B+w^o)$ (retirees consume only bonds)

Why Does Inflation Arise? The Monetary Rule

Flex-price world: Safe rate adjusts directly. No inflation needed.

$$1 = \beta(1 + r^*)\left[(1 - \chi) + \frac{\chi C^{w*}}{B' + w^o}\right] \quad \Rightarrow \quad r^*(B') > r^*(B_0) \quad \checkmark$$

Standard Taylor rule: r is pinned by π. The *only* way to raise r is through inflation.

$$r = \underbrace{\bar{\iota}}_{=r^*(B_0)} + (\theta_\pi - 1)\pi \quad \Rightarrow \quad r > r^*(B_0) \text{ requires } \pi > 0$$

Augmented rule: Intercept tracks $r^*(B)$. Bond market clears at $\pi = 0$.

$$r = \underbrace{\bar{\iota}'}_{=r^*(B')} + (\theta_\pi - 1)\pi \quad \Rightarrow \quad r = r^*(B') \text{ at } \pi = 0 \quad \checkmark$$

The inflation from fiscal expansion isn't caused by the fiscal expansion —
it's caused by the monetary rule refusing to accommodate a change in r^*.

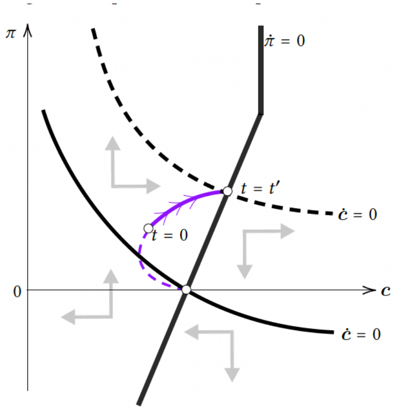
Debt Expansion Under the Standard Taylor Rule

Helicopter drop: government creates ΔB bonds, gives to workers for free.

Taylor rule: $r_t = \bar{\iota} + (\theta_\pi - 1)\pi_t\,, \quad \theta_\pi > 1.$

	Standard rule
Mechanism	$w\uparrow$
Bond clearing	Higher income
Inflation	Yes
Entrepreneurs	Lose
Pareto?	No

Bond mkt needs higher $r^*(B')$. Rule is rigid $\Rightarrow$ delivers r only via $\pi\uparrow$
Phillips curve: $\pi\uparrow$ requires $w\uparrow$
$\Rightarrow$ redistribution from entrepreneurs to workers.



GEE nullcline **shifts**; new SS at $(c', \pi' > 0)$

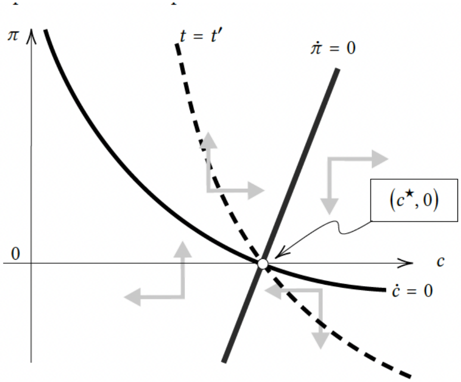
The Fix: Augmented Taylor Rule

Same fiscal expansion, but now $r_t = \bar{\iota} + (\theta_\pi - 1)\pi_t + \theta_b(B_t - B_0)$, $\theta_b = dr^*/dB > 0$.

	Standard	Augmented
Mechanism	$w\uparrow$	CB raises r
Bond clearing	Higher income	Higher return
Inflation	Yes	**None**
Entrepreneurs	Lose	Neutral
Pareto?	No	**Yes**

Augmented rule channels the expansion through the **price of savings** (r), not income (w).

No inflation $\to$ no deadweight loss, no redistribution.



GEE nullcline **rotates**; SS stays at $(c^*, 0)$

Comment 1: Micro-risks generate different losses

A new intra-generational cost under the standard rule:

- Inflation raises C^w, *widening* the worker–retiree gap
- Bond premium $bp = \chi\left(\frac{C^w}{B+\bar{w}^o} - 1\right)$ **rises**—partially undoes the insurance improvement
- First-order cost, hits **workers** (not entrepreneurs)

$\Longrightarrow$ The case for monetary reaction is **stronger**.

Comment 2: When Does $r^*(B)$ Increase? Not Always.

In AAA and CS formulations: more bonds $\Rightarrow$ higher r^*. **Always.**

But consider a 3-generation OLG (Y$\rightarrow$M$\rightarrow$O). The aggregate Euler is:

$$C_{t+1} = \beta(1 + r_{t+1})\big(C_t - c^o_t\big) + c^y_{t+1}$$

The wedge $\beta(1 + r)c^o - c^y$ now depends on the *distribution* (b^y), and agg B.

Key: who pays taxes to service the debt?

- Higher $r \Rightarrow$ higher taxes rB on workers
- Less disposable income $\Rightarrow$ may save *less* despite higher return
- If tax drag dominates: $r^*(B)$ *decreasing* $\Rightarrow$ no longer AAA's environment
  (cf. Mian, Straub & Sufi 2021—indebted demand)

Perpetual youth avoids this: taxes spread across all cohorts proportionally.

A richer OLG structure reveals: **who pays taxes determines whether bonds help or hurt.**

Comment 3: A Rising Tide Lifts All Boats?

With **demand-determined output**, the standard Taylor rule generates inflation **and** a boom:

$$\Pi' = \big(1 - w' - f(\pi')\big) \times Y' \quad \Longrightarrow \quad \text{share} \downarrow \;\; \text{but size of pie} \uparrow$$

- If size dominates share reduction: **even entrepreneurs gain**
- Pareto improvement possible **without** the augmented rule

Question for the Authors

Does the combination of Pareto criterion + exogenous output lead to an overly conservative prescription?

A utilitarian planner with endogenous output might prefer some inflation + a demand boom over zero inflation + pure redistribution.

Comment 3 (cont'd): What About Crowding Out?

Empirical fact: Safe asset share $\approx$ constant at $\sim$33% (Gorton, Lewellen & Metrick 2012).

When govt bonds expand, perhaps private safe assets contract?

AAA: no capital, no crowding out.
Debt expansion redistributes a fixed pie.

With capital: debt crowds out investment $\Rightarrow$ lower K', lower Y'.

- Acharya & Dogra (2022, JEEA): optimal debt trades off insurance vs. capital crowding out under dynamic efficiency
  but find: always optimal at ZLB to supply debt (AD channel dominates)
- AAA(2024, AER): micro-risks + investment

Thank you!

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- Cleanly separates distributional concerns from aggregate demand
- A natural addition to the teaching toolkit on monetary–fiscal interactions with heterogeneity

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SINK may float!

SINK vs. Micro Risks (AAA 2024, AER)

Same RPI criterion, different sources of inefficiency:

- **Micro Risks (2022):** Real model. Idiosyncratic risk, incomplete markets, potentially capital.
  - Debt improves welfare as an *insurance/liquidity* device
  - Pareto gains from better risk sharing even at constant K
  - $r < g + n$ is sufficient but not necessary — insurance margin provides gains even when $r > g + n$
- **SINK (2024):** NK model. No idiosyncratic risk, no capital. Perpetual youth.
  - Debt improves welfare through *intergenerational transfers* (Samuelson)
  - $r < g + n$ is both necessary and sufficient
  - **Adds:** monetary policy dimension (Taylor rule interactions)